

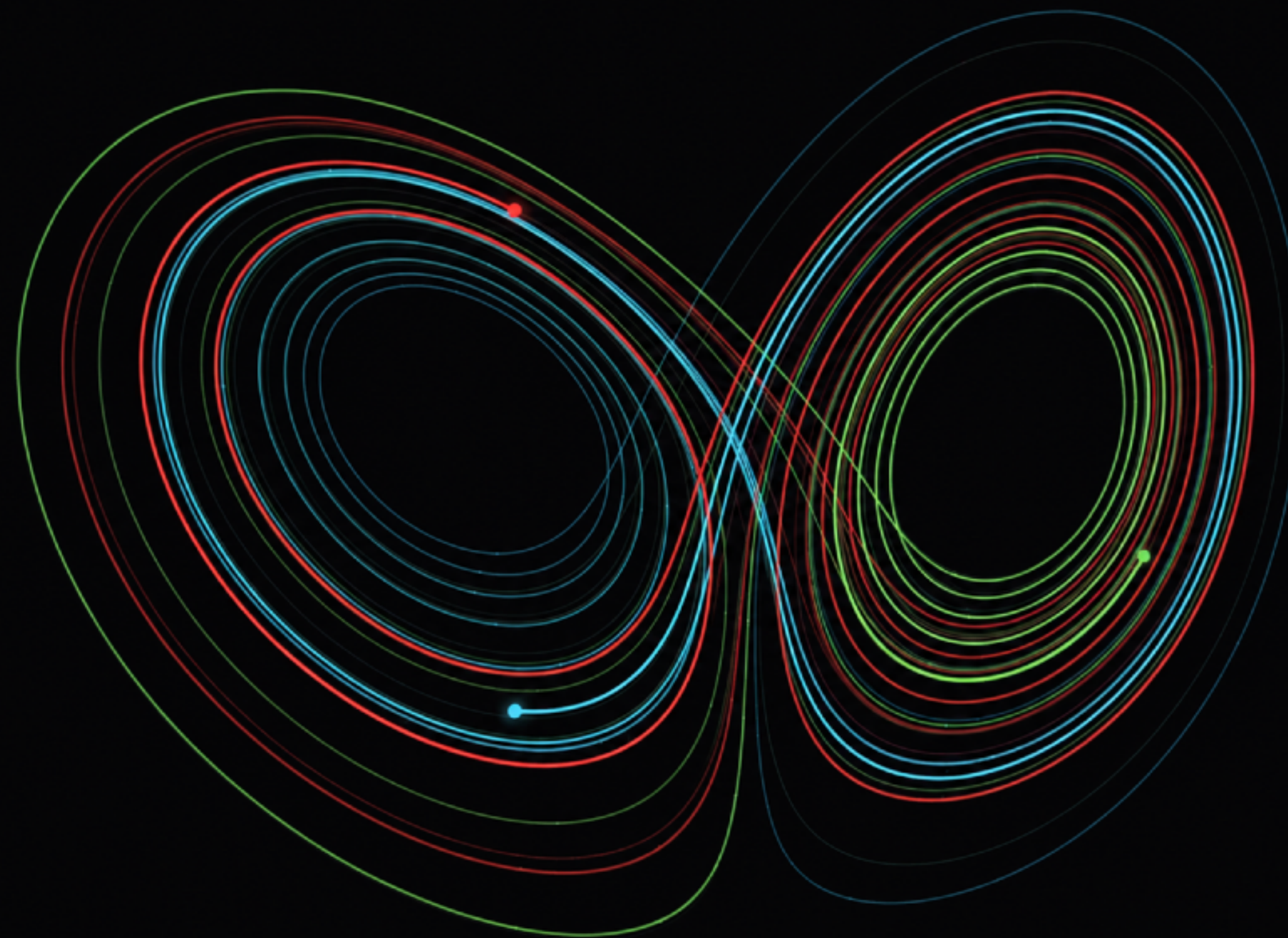
ICARM Summer School on Formalization:

Formalizing the Trapping Region Lemma for the Lorenz System in Lean

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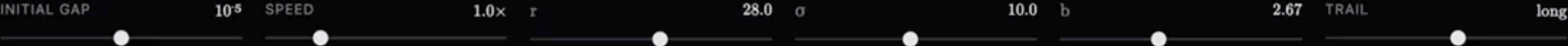


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- x 10.000000000 y 10.000000000 z 10.000000000
- x 10.00001000 y 10.000000000 z 10.000000000
- x 10.00002000 y 10.000000000 z 10.000000000



TIME

0.14

SEPARATION

1.59x10⁵

CHAOS THRESHOLD r_H

24.74

$r = 28.0 > 24.7$ — chaotic